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THE VOCAL REPERTOIRE OF THE BULLFROG (*RANA CATESBEIANA*)

by

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(With 8 Figures)

(Rec. 25-XI-1967)

INTRODUCTION

Vocalization is well developed in most species of frogs and toads. As BOGERT (1960) has suggested, anuran calls can be broadly classed into mating calls, territorial calls, release calls, warning calls, and distress calls. Of these different types of calls, the mating call has received the major attention. At least for some species, the mating call has been found to be species-specific and to provide a mechanism promoting reproductive isolation (MARTOF & THOMPSON, 1958; BLAIR, 1958). Inasmuch as the other types of calls are produced generally at lower sound levels and less frequently in the field, we have less information about the signal characteristics of these calls and the biological significance they convey in anuran sound communication.

In a recent study, I investigated the basis for the bullfrog's detection of the mating calls of other male bullfrogs as evidenced by his evoked vocal response. This study, involving terrarium colonies of bullfrogs maintained in a semi-natural environment in the laboratory, has been described in detail elsewhere (CAPRANICA, 1965, 1966). In the course of this research, it was found that the laboratory animals occasionally produced other types of calls. This report describes the types of calls typically produced and the significance that seemed to be associated with each call. Included in the discussion is the manner in which some of these sounds may be represented in the bullfrog's auditory system.

¹) Acknowledgements: Part of this work was conducted at the Center for Communication Sciences, Research Laboratory of Electronics, Massachusetts Institute of Technology. The M.I.T. Press kindly granted permission to reproduce some of the figures previously published in R. R. CAPRANICA, *The Evoked Vocal Response of the Bullfrog* (M.I.T. Press, Cambridge, 1965).

I especially wish to thank M. H. GOLDSTEIN, Jr., L. S. FRISHKOPF, and W. A. VAN BERGEIJK for their timely suggestions and discussions in the course of this study. The cooperation of M. J. MURRAY in implementing the laboratory is gratefully acknowledged.

METHODS

Studies of the vocal behavior of the bullfrog were conducted with two different colonies of bullfrogs kept in two specially designed terrariums. The terrariums were housed indoors in the laboratory in separate adjoining rooms. The temperature in each room was maintained between 67 and 72°F during the entire period of this study. An overhead lighting system, automatically turned on at the same time each morning (7:00 AM) and off each evening (8:00 PM), simulated day-night illumination cycles. A dim lighting level was maintained during the evening and early morning hours.

The terrariums were constructed in the form of large rectangular boxes: one terrarium measured $5\frac{1}{2} \times 3 \times 2\frac{1}{2}$ ft, and the other terrarium measured $8 \times 2 \times 3$ ft. The floor of each terrarium was filled to a depth of 4 to 8 inches with a mixture of sand and peat moss. A large pond, consisting of a plastic tank embedded in the sand and peat moss layer, could be externally drained and refilled with fresh tap water by means of underground tubing.

One of the terrariums, housing a colony of bullfrogs and decorated with artificial plants, is shown in Figure 1. The plants provided a more natural refuge and surrounding — without them the animals tended to remain in the ponds. The members of one colony were captured in their native habitat in New Jersey; the members of the other colony were captured in Wisconsin.

The behavior of the different members of each colony was observed by sitting quietly next to the terrarium for prolonged periods and simply noting the activity of the animals. At the same time, their vocalizations were recorded on magnetic tape by means of an omnidirectional microphone (University 70) suspended from the screened roof of each terrarium and a direct-recording tape recorder (Ampex 601). The details of the recording system, as well as the acoustical characteristics of the terrariums, have been described previously (CAPRANICA, 1965).

The gross temporal patterns of the recorded calls were studied by displaying them on a high-speed chart recorder (Offner pen recorder and Honeywell Visicorder). The temporal fine-structure within each call was analyzed by systematically displaying successive segments of the temporal waveform on a storage oscilloscope equipped with a calibrated variable delay in its horizontal sweep (Tektronix 564).

The spectral content within the different calls was measured with a Kay Sona-Graph audio spectrum analyzer. This analysis involved making first a normal spectrogram and then, to obtain a quantitative display of the amplitudes of the spectral components, amplitude cross-sections were taken

sequentially at several points along the time axis in the spectrogram (KERSTA, 1948; PULGRAM, 1959). All sections were made using an effective filter bandwidth of 45 cps. These measurements were later verified with an improved high-speed sound spectrograph developed at Bell Telephone Laboratories (PRESTI, 1966).

RESULTS

The vocal repertoire of the laboratory bullfrogs consisted of seven distinct types of calls. Each call could be easily identified either by listening to it or by its subsequent signal analysis. The degree of variation in call structure between different animals was relatively small to the extent that rarely was there ever any confusion (on my part) as to which type of call had been uttered by any of the laboratory animals. Furthermore, no significant differences between the New Jersey and Wisconsin bullfrogs could be found in the various types of calls or in the behavioral activity associated with their occurrence. As a consequence, the description which follows is considered as representative of the different types of calls recorded from the laboratory animals.

1. MATING CALL

The bullfrog's mating call, as in most anuran species, was made exclusively by the male. It consisted of a sequence of several similar croaks. The number of croaks in a call from any given male varied from 3-4 to as many as 14-15. The production of each croak was marked by a full inflation of the male's paired subgular vocal sacs. At the termination of a croak, the vocal sacs were rapidly deflated by shunting air back into the lungs to await the generation of the next croak. Individual croaks usually lasted from 0.6 to 1.5 sec and silent intervals between successive croaks ranged between 0.5 and 1.0 sec. A representative mating call is shown in Fig. 2.

The rise and fall times of the temporal envelopes of different croaks differed considerably. In some calls the rise times of the croaks were much sharper than the fall times; in other calls the fall times were much sharper; and in still other calls the rise and fall times were approximately equal. Based on an analysis of many different calls, a symmetrical envelope with rise and fall of 200-300 msec and overall duration of 0.8 sec seemed representative of an "average" croak within the bullfrog's mating call.

Although the gross temporal pattern of the mating call showed wide variations, the temporal fine-structure was more stereotyped. A pulsatile almost-periodic waveform having a repetition rate of approximately 100/sec



Fig. 1. One of the terrarium colonies of bullfrogs with which the studies of the bullfrog's vocal behavior were conducted.

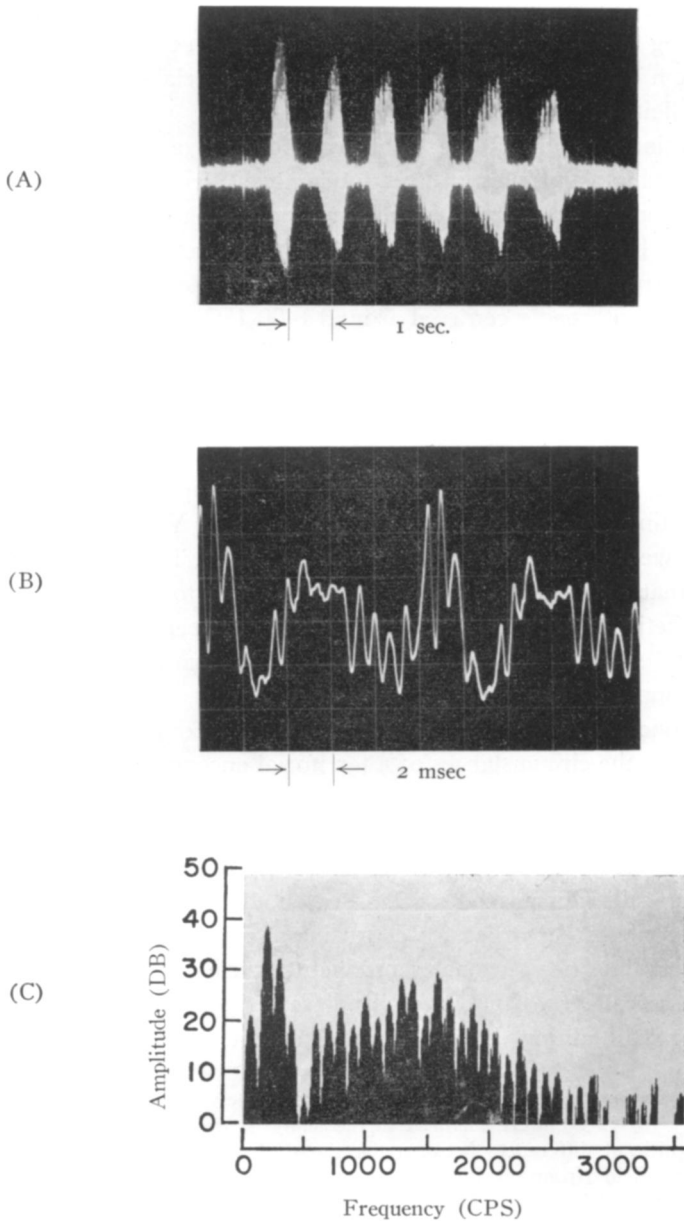


Fig. 2. A representative mating call from an adult male bullfrog. A The mating call, comprised of 6 similar croaks, in its entirety. B Expansion of a croak waveform showing the pulsatile wavelike shape in a single fundamental period (repetition rate approximately 100/sec). C Spectrographic section taken through the middle of a croak.

typified the fine-structure throughout each croak in the male's call (Fig. 2B). All of the bullfrog mating calls that I have analyzed, both those recorded in the field or from the laboratory animals ¹⁾, have invariably exhibited a waveform periodicity of 90-110/sec.

The almost-periodic nature of the croak waveform gave rise to a quasi-harmonic relationship (multiples of the fundamental frequency) between the spectral components ²⁾. The spectrum corresponding to the mating call of an adult male bullfrog (Fig. 2C) typically showed a concentration of energy in a low-frequency region centered around 200-300 cps and in a high-frequency region broadly centered around 1400-1500 cps. The envelope of the spectrum often had a relative amplitude dip in the mid-frequency region centered around 500-700 cps.

The significance of the anuran's mating call, at least in the case of the bullfrog, requires clarification. I have classed this call as a mating call on the basis that it is most frequently heard in the field during the height of the bullfrog's mating season. Males often call alternately. Yet this same call can be recorded well after the mating season has passed. In my laboratory study, I found that the males reliably called in response to the playback of natural and synthetic mating calls throughout a two-year period. Their vocal responses had the same signal characteristics as their mating call. Seldom did I notice attempts of amplexus, or other overt signs, associated solely with sexual behavior. When clasping did occur in the laboratory, it seemed to result more from the circumstances of a territorial encounter with another bullfrog (which will be described subsequently) than from an active intent to secure a mate.

These observations suggest a more general significance for the male bullfrog's mating call. During the mating season, his calling may indeed function to congregate other males for chorusing and possibly either attract a female or else initiate her seasonal reproductive cycle. At other times of the year, this same call may serve a totally different purpose (*e.g.*, species identification for maintaining the geographical extent and integrity of the

¹⁾ Field-recorded mating calls, when recorded at approximately the same temperature as in the laboratory (67-72°F), were included in the analysis for comparison purposes. I could find no significant differences between the signal characteristics of the mating calls recorded in the field (from several different geographical locations in the U.S.) and those recorded in the laboratory. Most of the field recordings were kindly supplied by the Cornell Library of Natural Sounds (Cornell University, Ithaca, New York) and by the Department of Herpetology, American Museum of Natural History (New York City).

²⁾ For a general introduction to the signal analysis of periodic, almost-periodic and aperiodic waveforms, the reader may refer to MARSHALL (1965) or MASON & ZIMMERMAN (1960).

local population). Therefore the classification of this call as having significance only for mating may be too restrictive.

2. TERRITORIAL CALLS

In maintaining the laboratory animals, it soon became evident that certain bullfrogs were more aggressive than others. In order to explore this behavior, the following routine was initially adopted. The animals were fed daily with small pieces of meat (beef heart and liver) dangled on the end of a fine thread.¹⁾ One of the most active males was singled out and purposely denied food. This was accomplished by dangling the meat nearby the male yet closer to a neighboring frog so that, even though the male oriented toward the food, another animal invariably took it first.

After several daily sessions of this procedure, the male gradually began to vocalize. His aggressiveness increased to the extent that, during a feeding session, he would attempt to drive all other animals away from his immediate vicinity. As these sessions continued, other members of his colony also began to vocalize and definite signs of territorial interaction appeared. That is, many of the laboratory bullfrogs tended to maintain exclusive locations in their terrariums. When another bullfrog happened to enter that location, a distinct call was often uttered by the proprietary frog. I found three different types of calls associated with this behavior. One type was made only by males, a second type was made only by females, and a third type was made by both males and females.

A. Male Territorial Call.

The territorial call of the male bullfrog seemed to serve as a challenge to other intruding bullfrogs that a territory was occupied and was being trespassed. If the intruder was a female, she typically remained silent and hopped away soon after the male had sounded his call. An intruding male also would often remain silent and quickly hop away. However, occasionally an intruding male would stand his ground and, in fact, sometimes answered the territorial call of the proprietary male with a similar call. In the event that the unwanted visitor did not retreat, the resident male invariably jumped at the intruder, often landing on top of him, and a "wrestling contest" ensued. This contest, involving leaps by each male directly upon the other, continued for several seconds or minutes until one or the other retreated.

¹⁾ Since the frog will not strike at a stationary target, the pieces of meat were dangled in a manner simulating a live insect. The thread, knotted at the end, was skewered through the beef heart or liver portion. If an animal chanced to take the food, the thread was released by a gentle tug.

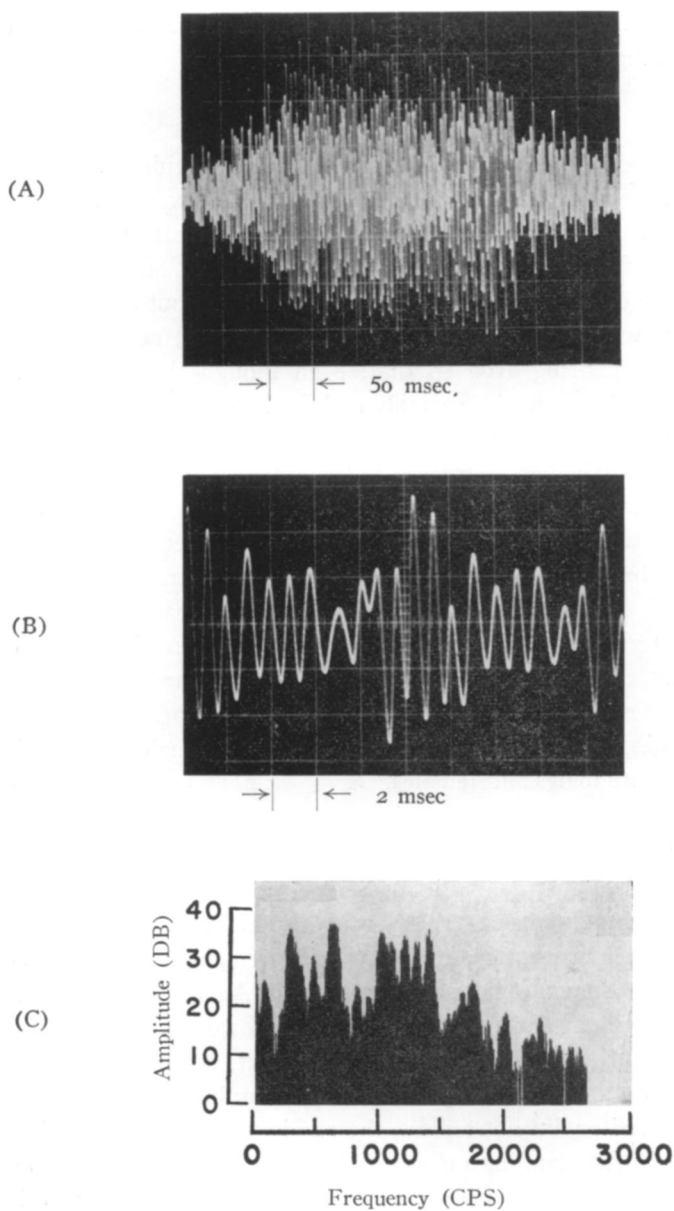


Fig. 3. Representative territorial call of a male bullfrog. A The call in its entirety. B Expansion of the call showing the repetitive structure in the waveform (repetition rate approximately 100/sec). C Spectrographic section taken through the middle of the call.

Usually each male would first issue his territorial call before leaping at his opponent: seldom did a male attack unless he had first uttered this call.

The male's territorial call is shown in Fig. 3. Its waveform, commonly lasting between 0.4 and 0.6 sec, has a quasi-periodic fine structure of approximately 100/sec. With the exception of a shorter average duration, the gross temporal envelope and quasi-periodicity of the fine structure in this call are quite similar to those of an individual croak in the mating call.

There were three features of the male's territorial call which reliably distinguished it from the mating call. First, whereas the mating call usually consisted of a sequence of at least three or more repeated croaks, there was only one isolated vocalization (represented by the waveform in Fig. 3A) ever given in this call; *i.e.*, it was never made repetitively. Second, the distribution of energy in the mating call of the adult male typically showed a low-frequency spectral peak in the region of 200-300 cps, whose amplitude clearly exceeded the amplitudes of the components between 500 and 700 cps (Fig. 2C). In comparison, the territorial call usually had a local spectral peak in the region of 500-800 cps with a relative amplitude equal to, or often greater than, that of the lower-frequency components (Fig. 3C). And third, the vocal sacs of the male were also involved in the production of his territorial call. However, in making this call, it appeared to me that he only partially inflated his vocal sacs (as compared with the full expansion, quite noticeable, in his production of the mating call). Such a partial inflation of the vocal sacs may account for the emphasis of spectral amplitudes in the region of 500-800 cps.

B. Female Territorial Call.

The female bullfrogs in the laboratory terrariums also maintained individual territories. Their call seemed to serve much the same function as the male's territorial call. The main difference in significance that I could detect was related to the sex of the intruding bullfrog. If the intruder was a male, he generally remained silent and hopped away soon after the resident female had uttered her call. Similarly an intruding female often would also remain silent and quickly hop away. However, sometimes an intruding female would not retreat but instead answered the territorial call of the resident female with a similar call. This response quickly led to a leap by one of the females directly upon the other in much the same manner as for contesting males, ending eventually in the retreat of one of the females.

The female's territorial call is shown in Figure 4. It consisted of a single vocalization with a duration generally between 1.4 and 1.8 sec. The gross temporal envelope of the call usually showed a gradual increase in amplitude

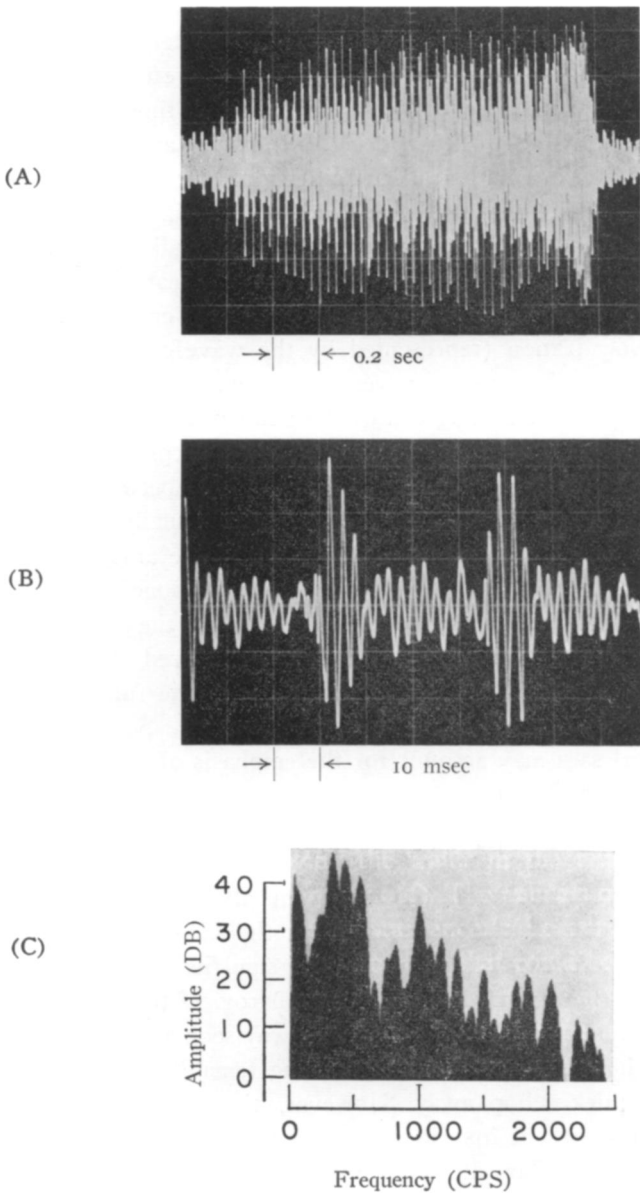


Fig. 4. Representative territorial call of a female bullfrog. **A** The call in its entirety. **B** Expansion of the middle of the call showing the repetitive structure in the major portion of the waveform (repetition rate approximately 28/sec). **C** Spectrographic section taken through the middle of the call.

during the initial 0.4-0.6 sec, relatively uniform amplitude during the remainder of the call, and then an abrupt termination with a short fall time (40-60 msec).

The waveform throughout the major portion of the call was quasi-periodic, quite pulsatile, having a repetition rate of 25 to 30/sec (Fig. 4B). The waveform during the termination of the call showed a very distinct change: it ended with a pronounced inflection in its periodicity. In referring to the representative call shown in Fig. 4A, the duration of this call was approximately 1.6 sec. The periodicity in the waveform throughout the first 1.4 sec was about 28/sec, as measured between successive pulsatile repetitions (Fig. 4B). During the last 0.2 sec of the call, the waveform periodicity increased monotonically from 28/sec to 90/sec (Fig. 5). This stereotyped change in waveform periodicity was a unique characteristic of the female's territorial call — no other bullfrog call showed a pronounced inflection in waveform periodicity.

During the terminal portion of the female's call, although the periodicity in the waveform changed, the spectral envelope of energy distribution remained approximately the same. That is, spectrographic sections taken over a single period anywhere in the call of Fig. 4A, including the last 0.2 sec, all showed similar energy distributions, as represented by Fig. 4C.

C. Male and Female Territorial Call.

A third type of call was made by both males and females. While the previously described male territorial call and the female territorial call were each given under rather stereotyped conditions, this third type of call could not be so reliably identified with a specific behavioral situation. However, inasmuch as its occurrence most often involved some sort of territorial interaction, I have classed this call, albeit tentatively, as a territorial call. I was unable to find any distinctive differences between the signal characteristics of the male's call and the female's call or in the circumstances under which they were given.

The call was most often made by the "victorious" bullfrog immediately following a territorial contest with another bullfrog (regardless of which animal had originally been the resident and which had been the intruder). It also was made, but less frequently, by a proprietary bullfrog when another member in the terrarium colony approached steadily from a distance, as if in anticipation of an intruder. If the approaching bullfrog did not change course, then this call ceased and, depending on the sex of the resident frog, one of the territorial calls described previously was given. Occasionally the call was given shortly after an extraneous disturbance in

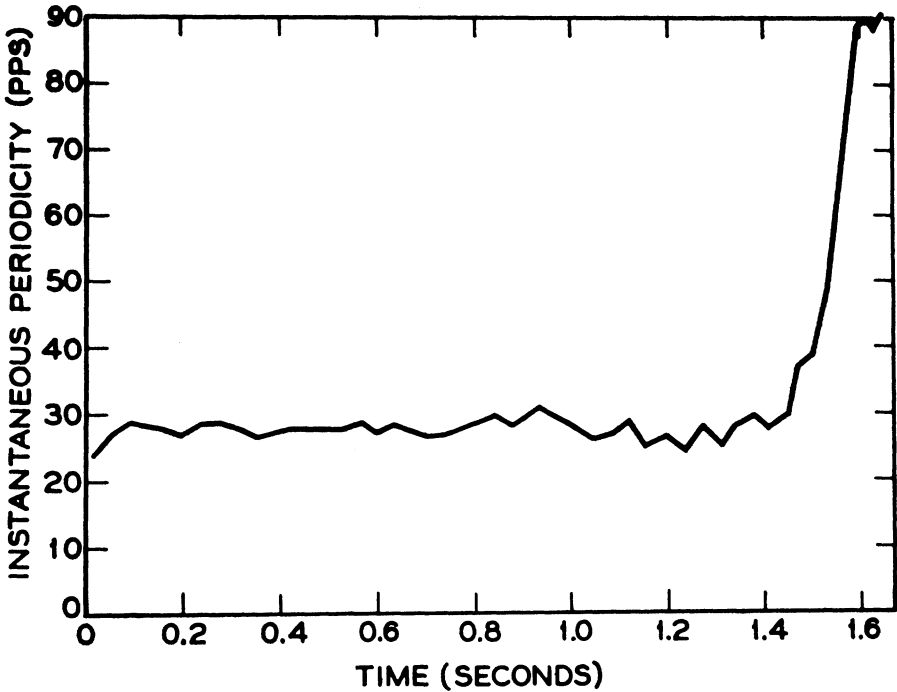


Fig. 5. Instantaneous periodicity, defined as the reciprocal of each fundamental period in the waveform, throughout the female's territorial call shown in Figure 4.

the laboratory (*e.g.*, changing the water in the terrarium ponds, sudden noises, turning room lights on or off). The animals often remained motionless for periods long after the disturbance had occurred, so that there were no overt signs of territorial interaction that could be related to this call in these instances.

A typical call, shown in Fig. 6, consisted of a single vocalization having a relatively symmetrical gross temporal pattern. It is uniquely characterized by a long duration, often between 3 and 6 seconds, and a quasi-periodic waveform with a very low periodicity, usually between 8 and 20/sec. The temporal fine-structure, being highly pulsatile, gave rise to a sound similar to a train of clicks. The call differed from the female's territorial call (Fig. 4) most notably in that its duration was longer, its waveform periodicity was much lower, and it did not show an inflection in periodicity during its termination. Inasmuch as the waveform was quasi-periodic, the spectral content remained essentially constant throughout the call (Fig. 6C).

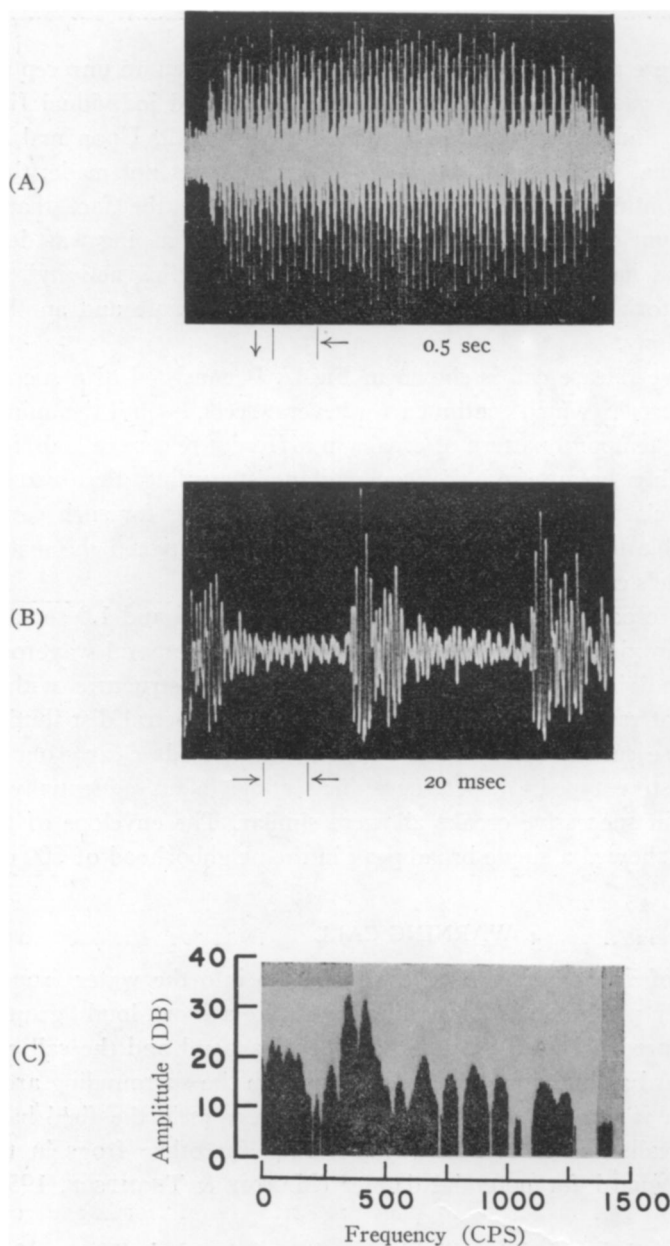


Fig. 6. Representative territorial call made by both male and female bullfrogs. A The call in its entirety. B Expansion of the middle of the call showing the repetitive structure in the pulsatile waveform (repetition rate approximately 12/sec). C Spectrographic section taken through the middle of the call.

3. RELEASE CALL

Release calls were made by both males and females. When an unreceptive female or a male was clasped by another male, the clasped individual first usually struggled and then would emit a characteristic call. Upon making this call, the animal was usually released. If this call was not made, then clasping could continue for a prolonged period (despite the fact that a considerable amount of struggling may have continued). Clasping was done only by males and most often occurred during the "wrestling activity" involved in a territorial encounter between a proprietary male and another member of his colony.

A representative release call is shown in Fig. 7. It consisted of a succession of repeated croaks which continued for several seconds until the animal was released. The temporal pattern of croaks in different release calls differed widely, possibly because of the concomitant struggling that usually accompanied the clasping of an unwilling animal. Allowing for such variability, I was unable to find any systematic differences between the male's call and the female's call.

Generally each croak in a release call lasted between 0.5 and 1.0 sec and was repeated approximately once every 1.5-2.0 sec. The temporal waveform comprising an individual croak usually had a pulsatile structure with a repetition rate that ranged between 60 and 85/sec. The spectral distribution of energy in those calls made by bullfrogs with their heads clearly out of water was rather stereotyped.¹⁾ Spectrographic sections taken sequentially at different points in successive croaks all were similar. The envelope of the spectrum usually showed a single broad peak in the neighborhood of 500 cps (Fig. 7C).

4. WARNING CALL

Many species of frogs, prior to an escaping leap into the water from a nearby bank, often produce a call sounding much like a short, loud "grunt". In the field one need only walk along the edge of a pond and the call can frequently be heard as individual animals jump from the surrounding areas into the water. In view of this escape behavior, observers in the field have associated these calls with a warning significance for other frogs in the vicinity to be on guard for potential danger (MARTOF & THOMPSON, 1958; BOGERT, 1960).

In the laboratory I found that individual male and female bullfrogs often

¹⁾ In many instances release calls were made by clasped animals while partly or totally submerged in one of the terrarium ponds. No attempt was made to record and analyze these calls.

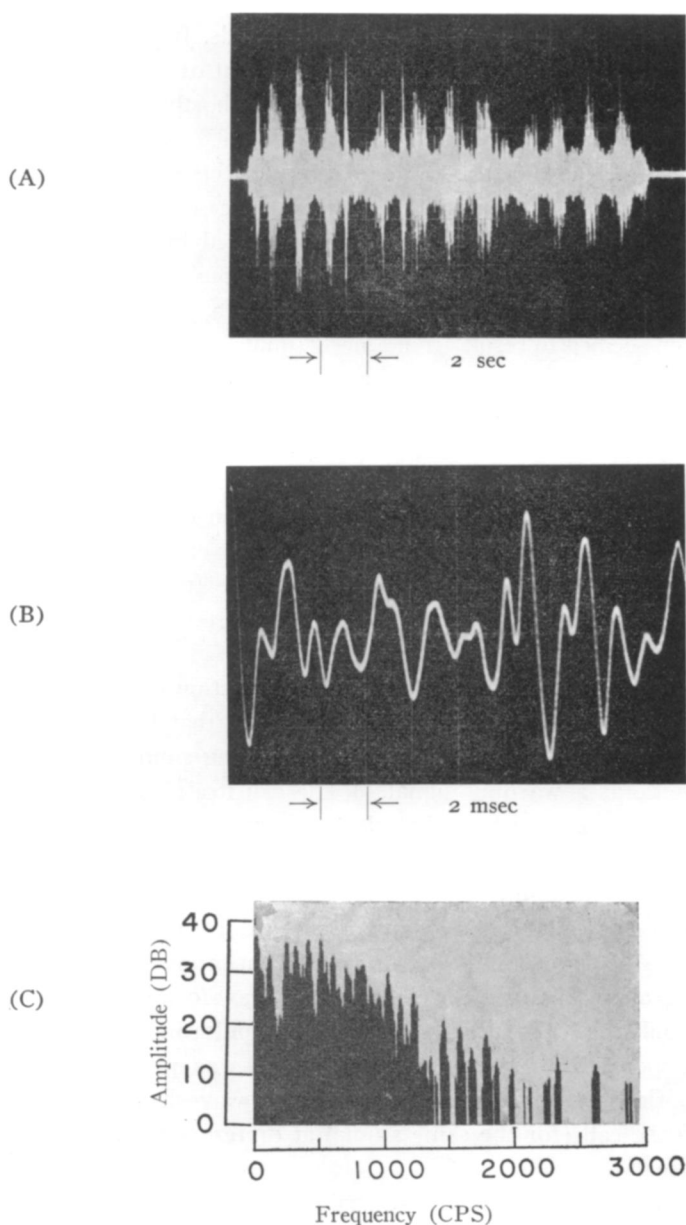


Fig. 7. Representative release call made by both male and female bullfrogs. A The call, comprised of a sequence of croaks, in its entirety. B. Expansion of the waveform in one of the croaks. C Spectrographic section taken through the middle of a croak.

produced short grunting sounds. Frequently an animal, upon making this call, followed with a leap into one of the terrarium ponds. Just as frequently, this same call was made by a bullfrog prior to a leap out of a pond. Another common occurrence of this sound was by a bullfrog shortly after issuing a different type of call, particularly a mating call by a male.

The biological significance of this call as having solely a warning function is perhaps too restrictive. Although these sounds were sometimes made by the terrarium animals if a sudden disturbance occurred in the laboratory, frequently they were made when there were no extraneous visual or auditory disturbances present. Rather than always involving a warning function, these sounds often seemed to result from a need simply for a rapid expulsion of air from the lungs. In the case of a laboratory animal jumping into or out of the terrarium ponds, deflation of the lungs provided a diving or landing advantage (*e.g.*, decrease the buoyancy of the diving animal). In those situations involving the male's mating call, upon termination of his call, he frequently reinflated his lungs to await the production of another mating call. Sometimes he kept his lungs fully inflated for several minutes before repeating his call. If he decided not to call again, since full inflation is not the normal lung volume for the bullfrog, he partially deflated his lungs and, when done rapidly, often a "grunt" occurred.

The assignment of a warning function to the production of this sound in all instances therefore seems questionable. It may be that this call, when followed by other stimuli such as the sight or splashing sound of a diving frog, *together* serve as a warning signal.¹⁾ The call itself may be but one component in a sequence of signals that alerts of approaching danger.

A representative call of this type from one of the laboratory bullfrogs is shown in Fig. 8. The temporal and spectral characteristics of this call are very similar to those of calls that are commonly heard in the field. The most distinctive features of this sound were its sudden onset and brief duration (generally lasting about 100 msec). Whereas the waveforms in all of the other types of bullfrog calls usually possessed a pulsatile quasi-periodic structure, the temporal fine-structure within this call was not repetitive and I was unable to find any systematic pattern in the waveshape of the fine structure during the call (for the same animal at different times as well as for different animals).

The spectral envelope of the energy distribution in this call usually had a broad peak below 1500 cps with a monotonic decrease in amplitude for in-

¹⁾ YERKES (1903, 1905) has suggested, by observing anuran behavior in the field and also by experimental studies of their reaction time to auditory stimuli, that the sound of splashing water is a highly significant stimulus for a frog.

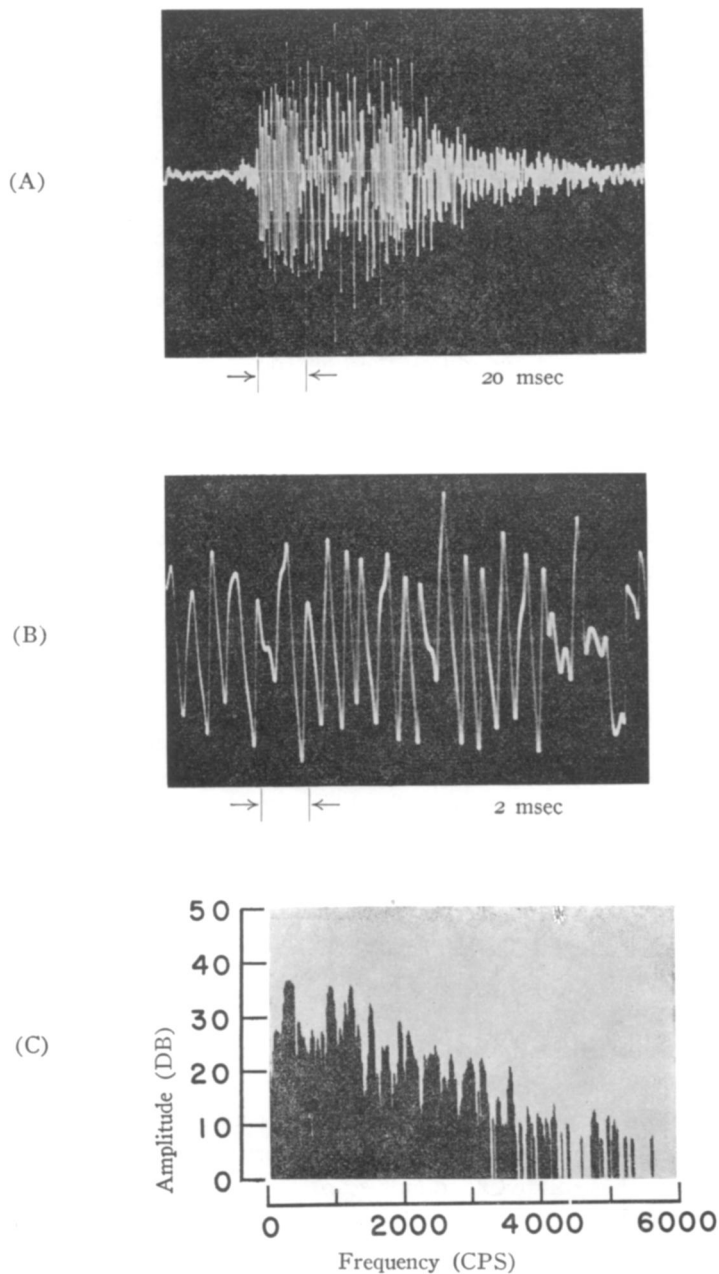


Fig. 8. Representative warning call made by both male and female bullfrogs. A The call in its entirety. B Expansion of the waveform in the middle of the call. C Spectrographic section taken through the middle of the call.

creasing frequencies. There were no significant differences between the male's call and the female's call.

5. DISTRESS CALL

The distress call is frequently made by an anuran in the field when he has been attacked by a mortal enemy. The significance of the call is thought to be for alerting other members of his species that an attack by a predator has occurred (BOGERT, 1960).

In the routine maintenance of the bullfrogs in the laboratory terrariums, I never observed a single instance of a distress call. However, in a related study involving an attempt to condition the bullfrog's cardiac rate to sound (CAPRANICA, 1965), I found that repeated application of electric shock, especially to the nasal region, would occasionally elicit this call. Unfortunately, the experimental arrangement at the time was not suitable for recording these vocalizations. I simply noted that the call was made by both sexes. Its most distinguishing features were its high pitch and its mode of production (made with the mouth wide open). The call sometimes lasted for several (1-5) seconds. After making this call, the animal usually relaxed into a very limp, passive state in what may have been an attempt to feign death.

There seemed to be little, if any, low-frequency energy in the distress call. Upon hearing it made in several instances by different bullfrogs in the laboratory, I would judge that most of the energy in the call is above 700-1000 cps. BOGERT (1960) has recorded the distress calls of three anuran species closely related to the bullfrog (*Rana grylio*, *R. pipiens*, and *R. clamitans*). Spectrograms of representative calls for these three species show distinct differences in their spectral patterns, thus suggesting that the call is species-specific. The dominant energy in the distress calls of each of the three species occurred above 1000-1500 cps.

DISCUSSION

The vocal repertoire of the bullfrog is limited to a small number of stereotyped calls. Each of these calls seems to be uniquely characterized by its temporal and spectral properties. Recent anatomical and electrophysiological studies suggest that the bullfrog's peripheral auditory system is well suited to detect and encode these different sounds.

The inner ear of the bullfrog contains two auditory sense organs: the amphibian papilla and the basilar papilla. FRISHKOPF & GOLDSTEIN (1963) have recorded from the primary nerve fibers (*i.e.*, from single units) that innervate these two organs in anesthetized bullfrogs. On the basis of sensitivity to pure tone bursts of various frequencies, they identified two distinct

types of single units. "Simple" units were most sensitive to frequencies between 1000 and 2000 cps.; "complex" units were most sensitive to lower frequencies, extending from vibration sensitive up to about 700 cps. The responses of all complex units to an excitatory tone could be inhibited by the addition of a second tone which, when presented alone, had little effect. Inhibition was graded and most effective when the frequency of the inhibitory tone occurred in the neighborhood of 500-700 cps. For total inhibition, the level of the inhibitory tone had to exceed the excitatory tone by at least 6 dB (regardless of the absolute sound intensity of the excitatory tone). In marked contrast to the inhibition of complex units, the simple units could not be inhibited by adding a second tone.

Both simple and complex units were capable of encoding temporal periodicity. Nerve spikes in both types of fibers were time-locked to trains of filtered pulses for pulse rates up to at least 100/sec; at 200/sec this synchronous "following" disappeared. Recently FRISHKOPF & GEISLER (1966) verified that each population of primary units derives from a separate papilla within the inner ear of the bullfrog: simple units from the basilar papilla and complex units from the amphibian papilla.

Except for the bullfrog's warning call (and possibly the distress call), the temporal fine-structure in the waveforms of all his other calls has a periodicity of 100/sec or less. Inasmuch as these repetition rates are within the limits of following by single fibers in the auditory nerve, the peripheral auditory system is capable of encoding and transmitting information about the temporal patterns in the different calls to the central nervous system.

Furthermore, each call has its spectral distribution of energy within the range of frequency sensitivity of the basilar and amphibian papillae. Thus the bullfrog's inner ear is well tuned to the spectral content of the sounds he produces. In fact, the tuning of the peripheral auditory system provides a considerable degree of "filtering" to enable the bullfrog to selectively detect the calls of other bullfrogs in his environment. For example, suppose we consider the spectral characteristics of his mating call necessary for evoking calling from another male bullfrog.

As we have seen (Fig. 2C), the spectral content in the croaks of a representative mating call from an adult male bullfrog often has a relative peak around 200-400 cps, a dip in amplitude around 500-700 cps, and then a broad distribution of energy centered around 1400-1500 cps. The location of the low-frequency spectral peak seems to be related to the size of the male and thus his age (CAPRANICA, 1965). That is, the calls made by young, juvenile males (body length 3-4 inches) often have their lower-frequency spectral peaks in the neighborhood of 500-700 cps, whereas the calls made

by large, adult males (body length 6-8 inches) have their spectral peaks located around 200 cps. Quite possibly the lower resonances associated with the larger vocal sacs of the older males are responsible for this downward shift in spectral energy.

Upon acoustic playback of these mating calls (CAPRANICA, 1965, 1966), I found that only the calls of *adult* male bullfrogs would reliably evoke calling from the male bullfrogs in the laboratory terrariums: the calls of *juvenile* males failed to evoke any vocal response (despite the fact that the laboratory colonies of bullfrogs included some of these same young males). When the playbacks of the adult mating calls were sharply (102 dB/oct) low-pass filtered at 500 cps or high-pass filtered at 1000 cps, then calling could no longer be evoked.

These results, verified in a systematic study of evoked calling to synthetic mating calls (CAPRANICA, 1965), have shown that the bullfrog's vocal response to the mating call of another bullfrog requires: (1) that the call contain sufficient energy both in the low-frequency region centered around 200 cps *and* in the high-frequency region centered around 1400-1500 cps; and (2) that none of the spectral components in the call's mid-frequency region centered around 500-700 cps have amplitudes greater than the spectral peak in the low-frequency region.

Thus it seems that there is a close correspondence between the frequency sensitivity of the bullfrog's inner ear and the spectral composition in his mating call. The fact that the mating call must have concomitant energy in the low- and high-frequency regions in order to evoke calling requires the excitation of the complex units from the amphibian papilla tuned to frequencies around 200 cps and excitation of the simple units from the basilar papilla tuned to frequencies in the neighborhood of 1400 cps. Since the frequency sensitivity of the bullfrog's inner ear apparently does not change with maturity (FRISHKOPF, 1965), the failure of the call from a juvenile male to evoke calling from other males, including other juveniles, can be attributed to the inhibition of their low-frequency complex units by the spectral peak around 500-700 cps in the young male's call. A maturing bullfrog will begin to evoke calling from other males when, due to the increasing size of his vocal cavities, the lower-frequency spectral peak in his mating call shifts downward from the (inhibitory) mid-frequency region into the (excitatory) low-frequency region. Note that the auditory basis for this mechanism, which evidently restricts the focus for calling in the field to adult male bullfrogs, has its origin in the filter characteristics of the peripheral auditory system.

These findings suggest that the spectral pattern of energy exciting the

two papillae in the bullfrog's inner ear is an important factor in determining the significance associated with a given sound. Thus we might expect that an acoustic stimulus whose energy is restricted primarily to the higher-frequency sensitivity of the basilar papilla, as in the distress call, would have a different significance than one whose energy is restricted to the lower-frequency sensitivity of the amphibian papilla. Several behavioral observations support this conclusion.

YERKES (1903) found that the presentation of "shrill", high-frequency sounds (Galton and Appunn whistles) produced a noticeable decrease in the respiration rate of the green frog (*Rana clamitans*). Low-frequency sounds (100-500 cps tuning forks) had little or no effect on the respiration rate. YERKES concluded that the decrease in respiration rate to high-frequency sounds was due to fright and, in view of the predominance of high-frequency energy in the green frog's distress call, reflected a fear significance associated with this call.

While a low-frequency sound by itself had little effect on respiration rate, its accompaniment with a small, moving visual target produced a greater increase in the green frog's respiration rate than to the moving target alone. YERKES suggested that this increase might be related to the low-frequency sounds made by a flying insect and, therefore, low-frequency sounds alert a frog to the presence of a small moving target.

The recent experiments of BIRSNER & MELZACK (1966) corroborate the finding that higher-frequency sounds have a fright significance for the frog. They found that a tone of 1500 cps would inhibit the approach response of the bullfrog to a moving object. In the absence of the tone, bullfrogs would approach moving discs of much larger size than when the tone accompanied the presentation of the moving targets.

In my studies of the evoked vocal behavior of the laboratory bullfrogs, I found that the occasional presentation of repetitive tone bursts between 600 and 2500 cps at moderately high intensities (90-100 dB SPL) would not evoke calling but, instead, occasionally produced a fear response: the animals (both males and females) immediately jumped into the terrarium ponds.¹⁾ That is, prior to the presentation of the tone bursts, only a very few bullfrogs were normally in the terrarium ponds at any given time — most of the animals maintained their favorite terrestrial territories. Soon after the tonal presentation began, the animals left their territories and "escaped" into the water: within 15-20 seconds most of them had already entered the

¹⁾ This behavior could be elicited more often during the evening hours when the rooms in which the terrariums were kept were dimly illuminated.

ponds. They remained in the water and made repeated efforts to submerge as long as the tone bursts continued. Upon termination of the tone bursts, the animals slowly crawled out of the ponds and resumed their territories. As I have mentioned earlier, most of the energy in the bullfrog's distress call seems to be above 700 cps. It may be that the escape behavior of the laboratory animals to the higher-frequency tone bursts is related to the spectral content and significance of this call.¹⁾

In marked contrast to the escape behavior of the laboratory bullfrogs to the higher-frequency tones, the presentation of tones of approximately 400 cps or lower would not elicit this behavior: the animals made no movements and remained in their territories. Furthermore, while a high-frequency tone by itself initiated escape behavior, its combination with a low-frequency tone would not. Instead, the combination of a tone of 200-300 cps with a tone of 1300-1500 cps, when presented in repetitive croaklike bursts, frequently evoked calling from the laboratory male bullfrogs. This was the simplest approximation to the spectral characteristics of the mating call that would evoke calling. If, to this combination, a 500-cps component of sufficient amplitude were added, then calling was suppressed.

These behavioral observations indicate that the bullfrog's response to a high-frequency stimulus, which excites primarily his basilar papilla, is quite different from his response to a low-frequency stimulus, which excites primarily his amphibian papilla. Further, a stimulus which excites both these organs simultaneously can elicit still another behavior (namely evoked calling, providing there is insufficient energy in the inhibitory mid-frequency region of the amphibian papilla). It would seem, therefore, that the spectral characteristics of each of the calls in the bullfrog's vocal repertoire, and of other meaningful sounds in his environment, must be highly important in specifying their significance.

This is not to imply that the temporal characteristics of these sounds are unimportant. In my studies of the bullfrog's evoked vocal response, I found that those synthetic mating calls which would evoke the most calling from the laboratory males had their waveform periodicities in the neighborhood of 100/sec. For a given spectral envelope, the degree of calling to these stimuli decreased as the periodicity in their waveforms was lowered to 50/sec; and calling could no longer be evoked when the periodicity was

¹⁾ However, YERKES (1905) and BOGERT (1960) have observed instances of an *anuran* uttering a distress call in the field. They claim that other frogs in the immediate vicinity made no overt movements in response to the call, and therefore suggest that the call may serve to alert other frogs but that visual stimuli probably predominate in initiating a frog's escape behavior.

lowered to 25/sec. Inasmuch as the periodicity in the waveform of the natural mating call is about 100/sec (Fig. 2B) while in the waveforms of two of the territorial calls (Fig. 4B and 6B) it is below 30/sec., calling may have ceased as the waveform periodicity was lowered because these stimuli took on a different behavioral significance — *e.g.*, a territorial significance in which a vocal response is not normally given.

It therefore seems unlikely that the bullfrog's discrimination of the different calls in his repertoire rests solely on their spectral characteristics. Inasmuch as each call seems to possess distinctive temporal features, such as waveform periodicity, duration, and overall gross pattern, these temporal features are probably of considerable importance to the bullfrog in his recognition of these sounds. Further analysis of the spectral and temporal characteristics of these sounds and the behavioral significance attached to them may well elucidate the important properties of acoustic stimuli involved in anuran sound communication.

CONCLUDING REMARKS

This study has been conducted with bullfrogs maintained in captivity. The advantages of a laboratory study are that the animals can be observed at close range for prolonged periods and their vocalizations can be recorded without contamination by extraneous natural sounds. While an attempt was made to provide a semi-natural environment for the animals, nevertheless these laboratory conditions were but approximations to the bullfrog's native habitat. The question therefore arises as to how representative has the bullfrog's behavior in the laboratory been of his behavior in the field.

Three of the calls recorded in the laboratory have also been recorded in the field: mating call, release call, and warning call. The signal characteristics of these field-recorded calls are very similar to those recorded in the laboratory. Further field studies are needed to confirm the laboratory observations regarding the other calls in the bullfrog's vocal repertoire.

SUMMARY

1. The vocal repertoire of bullfrogs maintained in laboratory terrariums in a semi-natural environment consisted of a distinct set of stereotyped calls. Each call could be readily identified with specific patterns of behavioral activity, thus indicating the significance represented by each of these sounds.
2. Mating calls were made only by males. The call by an adult male often evoked calling from other male bullfrogs; the call by a juvenile male did not evoke calling.
3. Territorial calls consisted of three distinct types: (i) one type made only by males, (ii) a second type made only by females, and (iii) a third type made by both males and

females. These calls seemed to alert other bullfrogs that a territory was occupied and was being trespassed.

4. Release calls were made by both sexes. When an unreceptive female or a male was clasped by another male, the clasped animal first usually struggled and then emitted this call, whereupon it was released.

5. Warning calls were made by both sexes. This call was often made prior to a leap into or out of the terrarium ponds. It also was sometimes made by a bullfrog shortly after having produced one of the other calls in his vocal repertoire.

6. Distress calls, made by both sexes, could be evoked by repeated application of a noxious stimulus. This high-pitched call was always made with the mouth wide open.

7. The bullfrog's peripheral auditory system is well suited to detect and encode the calls in his vocal repertoire.

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ZUSAMMENFASSUNG

1. Unter möglichst natürlichen Bedingungen im Laboratorium gehaltene Ochsenfrösche verfügen über eine Anzahl unveränderlicher Laute. Jeder von ihnen gehört zu bestimmten Verhaltensweisen, wonach sich seine Bedeutung festlegen lässt.

2. Nur ♂♂ äussern den Paarungsruf; ansteckend wirkt er nur aus der Kehle ausgewachsener ♂♂.

3. Dreierlei Revierverteidigungsrufe sind zu unterscheiden: einen äussern nur ♂♂ den zweiten nur ♀♀, den dritten beide Geschlechter. Alle scheinen in fremde Reviere eindringende Artgenossen zu verwarnen.

4. Befreiungsschreie lassen beide Geschlechter hören. Nicht paarungsbereite ♀♀ und jedes ♂ wehren umklammernde andere ♂♂ ab; wenn sie danach diesen Schrei ausstossen, kommen sie meist frei.

5. Den "Warnschrei" rufen beide Geschlechter oft, ehe sie ins Wasser springen oder an Land gehen, manchmal auch folgend auf irgend einen anderen Laut.

6. Sehr hohe Hilferufe aus weit geöffnetem Mund äussern beide Geschlechter nach wiederholter peinlicher Belästigung.

7. Der Gehörapparat des Ochsenfrosches ist daran angepasst, diese Artlaute nach ihrer Bedeutung zu unterscheiden und genau zu orten.
